

Vision-Based Flow-Front Assessment in Vacuum-Assisted Resin Transfer Molding

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Vacuum-Assisted Resin Transfer Molding (VARTM) operators read the visible advance of the resin flow front under the transparent vacuum bag as their primary process indicator. Prior vision systems for VARTM and Liquid Composite Molding use the same classical visual primitives, namely reference-frame differencing, Otsu thresholding, and morphological cleanup, but each system uses a different subset, no system reports the joint and individual contribution of those primitives, and no labeled video benchmark exists against which to attribute that contribution. This work presents an integrated classical computer-vision pipeline that combines a peak-brightness reference, a darken-only difference, region-of-interest restriction, dynamic-lag reference selection, morphological cleanup, and persistence-based temporal locking, evaluated on a fifty-five-frame hand-labeled subset spanning eleven distinct VARTM infusion runs. The integrated configuration reaches mask Intersection-over-Union of $X.XXX$ (95 % bootstrap confidence interval $[a, b]$) and boundary F_1 of $Y.YYY$ (CI $[c, d]$). A component-removal ablation shows that disabling any single primitive drops mean IoU by at least ΔIoU absolute, with the peak-reference and temporal-lock components accounting for the largest individual deltas. The pipeline runs at thirty frames per second on a single CPU and at K frames per second on a CUDA implementation across the eleven samples.

I. Nomenclature

F_t	=	converted-colorspace frame at time index t
G_t	=	reference image at time index t
G_t^*	=	effective reference (peak map when peak-reference enabled, otherwise channel-zero of G_t)
P_t	=	per-pixel peak-brightness map at time t
D_t	=	per-pixel response field at time t
R	=	binary region-of-interest mask
H, W	=	frame height and width in pixels
C	=	channel count of the working colorspace
α	=	exponential-moving-average factor for running-reference mode
κ	=	sqrt-area growth factor estimated from calibration frames
ρ	=	target wet-fraction for dynamic reference mode
λ	=	user lag scale factor for dynamic reference
τ_t	=	elapsed time at frame index t in seconds
$\Delta\tau_t$	=	dynamic lag in seconds for the reference frame
w_c	=	channel weight in the delta computation
N_{calib}	=	number of calibration frames before dynamic reference activates

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f	=	video frame rate in frames per second
IoU	=	Intersection-over-Union mask agreement score
F_1	=	boundary F_1 score, mean across pixel tolerances $\tau \in \{1, 3, 5\}$

II. Introduction

VACUUM-Assisted Resin Transfer Molding (VARTM) is the dominant process for one-off composite laminates from research panels through small-series aerospace structural parts, and the principal observable signal available to the operator during a run is the visible advance of the resin flow front under the transparent vacuum bag [1]. The position, shape, and rate of that front carry process-quality information that other, more instrumented modalities recover only indirectly. Distributed dielectric arrays [2] and area-sensor grids [3] produce excellent local wetness measurements but require sensors embedded in the lay-up, and fiber-optic frequency-domain reflectometry [4] demands instrumentation no operator on a research bench will install for a one-off panel. Direct video of the bag is the one modality that costs nothing extra, sees the entire laminate at once, and is already recorded in most labs as a procedural artifact.

Prior camera-based VARTM and Liquid Composite Molding (LCM) systems agree on the basic recipe. Difference the live frame against a reference image of the dry preform, threshold the response into a binary candidate mask, and clean the mask with morphology [5,6]. They disagree on every detail that controls how well the recipe works. Some systems clip the difference so that only darkening counts as wetting and others do not [6]. Some track a per-pixel running maximum as the reference and others pin a single early frame [5]. Some impose a temporal-locking rule that holds a positive detection through transient dropouts and others operate frame-by-frame [7]. None of these systems reports the joint contribution of the components it uses, the marginal cost of each component it discards, or the agreement of the resulting mask against a labeled video benchmark, because no shared labeled benchmark exists. The practitioner is left to take the recipe at face value.

The work is hard for three concrete reasons that have to be addressed by anything claiming to do it. The vacuum bag produces strong specular reflections from lab lighting that look bright in every channel and shift on every operator move. The bag wrinkles and creases that pre-date the front are static dark features that do not represent wet fabric and must not be mistaken for it. The lighting itself drifts across a multi-minute infusion as lamps warm up and the operator moves equipment. A pipeline that fails to address any one of these will mark unwetted fabric as wet on most frames of most infusions. Prior camera-based systems individually defeat one or two of the three. None reports which component defeats which failure mode, and none reports the price paid for leaving any of them out.

This paper presents an integrated classical-computer-vision pipeline that combines, in one configuration, the visual primitives the prior systems use in isolation. The pipeline restricts the comparison to a region of interest, projects the working frame into the lightness channel of CIELAB, accumulates a per-pixel peak-brightness reference that absorbs lighting drift, computes a darken-only difference that rejects specular highlights, dynamically lags the reference frame against fill rate, thresholds the response with Otsu, cleans the mask with morphological closing and opening followed by connected-components filtering, and locks pixels whose wet label has persisted for several consecutive frames. On a 55-frame hand-labeled subset spanning eleven distinct VARTM infusion runs, the integrated configuration reaches mask Intersection-over-Union (IoU) of $X.XXX$ with a 95 % bootstrap confidence interval of $[a, b]$ and boundary F_1 of $Y.YYY$ with confidence interval $[c, d]$. A component-removal ablation on the same subset shows that disabling any one of the named components drops mean IoU by at least ΔIoU absolute, with the peak-brightness reference and the temporal lock accounting for the largest individual deltas. The pipeline runs at 30 frames per second on a single central-processing-unit (CPU) thread, and a CUDA implementation reaches K frames per second on the same eleven samples.

The contributions of this paper are the following.

1. An integrated VARTM flow-front segmentation pipeline that combines peak-brightness reference, darken-only difference, ROI restriction, dynamic-lag reference selection, Otsu-plus-offset thresholding, morphological cleanup, and persistence-based temporal locking in one configuration, with mean IoU $X.XXX$ and boundary

F_1 Y.YYY reported with 95 % bootstrap confidence intervals on a 55-frame hand-labeled subset spanning eleven distinct infusion runs.

2. A component-removal ablation showing that every named primitive contributes a measurable IoU delta to the integrated configuration, with the peak-brightness reference and temporal lock contributing the largest individual deltas. The ablation is the empirical evidence behind the claim that the joint composition matters.

Section 2 places this work among the three families of prior work that touch the problem. Section 3 walks each component of the pipeline and points forward at the ablation evidence. Section 4 documents the labeling protocol and the eleven-sample dataset. Section 5 reports the integrated configuration’s agreement against the labeled subset and the per-component ablation. Section 6 reports CPU and CUDA throughput. Section 7 discusses failure modes and design tradeoffs. Section 8 concludes.

III. Related Work

Prior work on Vacuum-Assisted Resin Transfer Molding (VARTM) and the broader Liquid Composite Molding (LCM) family touches the flow front from three different directions. A first body of work uses flow-front information to act on the process through closed-loop control or fault response. A second body observes the process to characterize it, typically with embedded sensors or auxiliary cameras for offline study. A third body models the underlying Darcy physics or inverts it from sparse measurements. VRIFA sits at the seam of all three because it produces the per-frame full-field flow-front mask that the first family assumes is available, that the second family currently approximates with sensor grids, and that the third family typically receives only as a handful of digitized contours from a manual annotation pass. The remainder of this section addresses each family in turn and closes with the specific gap that VRIFA fills.

A. Use: vision and sensors that act on the VARTM process

Closed-loop VARTM controllers have a long lineage. Multi-segment injection-line work coupled point sensors with a finite-element flow model so that valves could be re-sequenced in real time to chase the flow front toward starved regions and away from race-tracks [8]. The control loop is genuinely real time and the empirical results are convincing for plate geometries, but the feedback signal is a sparse vector of resin-arrival times at instrumented locations, which means dry-spot detection only works when a sensor happens to lie inside the dry region. VRIFA differs by treating every pixel inside the bag as a potential sensor, so dry-spot evidence is not gated on instrumentation density.

A more recent line of work replaces point sensors with a top-mounted camera. Mesogitis and co-workers built a webcam-plus-microcontroller rig that segments the resin flow front with classical machine-vision routines and toggles inlet valves on the resulting front position [5]. Mendizabal et al. close a similar loop on resin film infusion, regulating flow-front velocity to a fixed setpoint with a PID over an OpenCV pipeline and reporting tensile-strength gains of up to 18 percent at the optimal velocity window [6]. Both systems demonstrate that vision alone can drive VARTM control, but neither reports the joint or individual contribution of the segmentation primitives they use, neither evaluates the segmentation against a labeled video benchmark, and both treat the segmentation step as a black box that succeeds or fails as a unit. The work in this paper differs by combining the same family of primitives in one configuration and reporting the agreement of the joint configuration and the IoU cost of disabling each component on a labeled subset spanning eleven distinct infusion runs.

Adjacent to control are vision-based defect-detection systems for composites more broadly [9]. These pipelines now lean almost entirely on convolutional or transformer detectors trained on hand-curated defect crops, and a recurring bottleneck across that survey is the cost of pixel-accurate labels. The work in this paper is upstream of that bottleneck. The integrated classical pipeline produces per-pixel front masks deterministically and characterizes which components are responsible for which IoU points, which gives a downstream learned detector [10,11] a labeled-data starting point whose error structure is decomposable rather than monolithic.

B. Observe: sensor-based and vision-based process characterization

A second family treats the flow front as something to be measured rather than acted on. Konstantopoulos and co-authors review the in-line sensing landscape and group the field around dielectric, thermal, ultrasonic, fiber-optic, and electrical-resistance sensors [1]. Dielectric tracking by Yenilmez and Sozer and others gives unsaturated and saturated front positions at each electrode pair, with an installation cost that is hard to beat, but the spatial resolution of the

resulting front is limited by the electrode pitch [2]. Matsuzaki and co-workers dramatically improve coverage with an area-sensor array that tiles a polyimide film into a dense capacitive matrix and recovers the impregnated-area ratio per cell, which is the closest sensor-side analogue to a full-field flow-front mask [3]. The resulting front is, however, still a quantized grid that is destructive to instrument inside production tooling and that adds permeability to the stack-up.

Fiber-optic strategies use Fiber Bragg Grating arrays or Optical Frequency Domain Reflectometry to time-stamp resin arrival at every grating along an embedded fiber, and recent distributed-sensing implementations push toward life-cycle structural health monitoring of large CFRP parts [4]. These methods deliver excellent temporal precision and can survive infusion and cure, but the same coupling between sensor density and front fidelity reappears, since arbitrary front shapes between gratings are interpolated, not measured. A small but growing thread fuses dielectric and visual data with learning-based fusion. The recent AI-based dielectric-plus-visual monitor by Esposito and co-workers detects and tracks the front with a YOLO detector trained on labeled video and uses dielectric channels as a redundant signal [7]. That pipeline is closest in spirit to the front producer described here, but it is reported on a single-fabric study, it does not report which preprocessing components carry which IoU points, and the detector is trained rather than classical. The work in this paper differs by combining the classical primitives the prior systems use in isolation, evaluating the joint configuration on a labeled subset spanning eleven infusion runs, and reporting the marginal IoU effect of each component.

Stieber and co-workers show that learning a flow-front “image” from a sparse pressure-sensor grid is a tractable surrogate task and report time-step accuracy near 92 percent on simulated injection runs [12]. A follow-on extends the same approach to material-property inference under a simulation-to-real transfer pretext and shows IoU around 0.50 with transformer backbones [13]. These are excellent demonstrations of what a network can learn given ground-truth flow-front maps. They presume the maps exist, and they obtain them from FEM injection simulations rather than from real video. VRIFA addresses the upstream half of that pipeline by emitting the flow-front masks from real VARTM runs that such networks need for fine-tuning.

C. Model: physics-based forward and inverse flow-front formulations

The third family models the front analytically or numerically. Forward Darcy-flow simulators for VARTM now routinely couple a depth-averaged Darcy solver to a level-set front with high-permeability-medium layers and progressive-saturation effects [14], and end-to-end VARTM simulation with reinforcement compaction and post-filling pressure relaxation has been validated experimentally for plate and stiffened-plate geometries [15]. These models predict the flow front given a permeability field and a boundary condition. The inverse direction is harder. Comas-Cardona and co-authors recover an in-plane permeability field by coupling optical areal-weight measurement with central-injection front observation, and similar work by Di Fratta, Klunker, and Ermanni inverts pressure-sensor traces against an analytic front model to update injection parameters online [16,17]. These methods are mathematically careful, but they all depend on having the front geometry as input. In Comas-Cardona that geometry comes from a manually thresholded plate experiment, and in the sensor-based works it comes from arrival-time interpolation. VRIFA lowers the cost of populating that input by emitting per-frame, per-pixel front masks directly from the production video stream that is already being recorded for documentation.

Recent work has begun to merge model and learning. Park and co-authors couple electromechanical PZT signals to a tree-plus-GAN model that both identifies the live front and forecasts its near-future position [18], and physics-informed neural permeability inversion has been explored in synthetic Darcy benchmarks [19]. Both are promising for closed-loop VARTM, and both inherit the same dependency on a labeled flow-front signal during training. Without an open, retargetable source of that signal in real video, every group reproducing or extending these methods has to instrument a new bench.

D. Building blocks: classical CV foundations

VRIFA’s segmentation core stays deliberately within well-understood classical operators. Otsu’s bimodal threshold provides the wet-versus-dry separation in each frame [20], the Suzuki-Abe topological-border-following algorithm provides hierarchically consistent contour extraction over the resulting binary [21], and the Kim et al. codebook background model handles the slow illumination drift typical of multi-hour infusions without retraining [22]. None of these primitives is novel. The contribution is in their composition, in the per-stage parameterization for VARTM video,

and in releasing the result as a Rust binary that another laboratory can run on its own footage. The recent material-defect-detection survey notes that the absence of openly retargetable preprocessing pipelines is a primary bottleneck for deploying deep models in composite quality control [9], and modern detectors such as YOLO illustrate why that bottleneck matters because they are starved for in-domain pixel labels rather than for capacity [23].

E. Where this work fits

The three families above all touch the flow front, but none of them reports the question this paper answers. The control family treats segmentation as a black box and reports downstream control quality, not segmentation quality. The observation family reports excellent agreement at sensor electrodes and gratings, but interpolates between them rather than measuring per-pixel front geometry from video. The modeling family assumes a per-pixel front exists and consumes it, which is precisely the artifact the upstream segmenter must produce. Within the camera-based subset of the observation family, prior systems differ in which subset of the available primitives they use and report only end-to-end behavior on a single mold. None reports the joint contribution of the components in use, none reports the marginal cost of any component left out, and none evaluates against a labeled video benchmark spanning more than one mold. This paper supplies that evidence on eleven distinct infusion runs and on a 55-frame hand-labeled subset.

IV. Method

A. Pipeline overview

The detection algorithm treats each video frame as a comparison against a quiescent reference image of the dry preform. Where resin has wetted the fabric, the local appearance darkens, and the absolute change is largest near the advancing flow front. The pipeline first checks for camera motion against the previous frame and, when motion crosses a configured threshold, registers the live frame back into the reference coordinate system before any difference is computed. The registered frame is then optionally smoothed with a pre-delta blur whose output feeds both the running peak map and the difference computation, so the reference and the comparison see the same bandwidth-limited input. The pipeline then computes the per-pixel difference inside a rectangular region of interest, clips it to admit only darkening so that specular highlights from the vacuum bag are rejected, normalizes the resulting field, thresholds it into a binary candidate mask, cleans the mask with morphological closing and opening, removes small connected components, and finally locks pixels whose wet label has persisted for several consecutive frames. The locked mask is the canonical output, the optional heatmap renders the underlying delta field for visual diagnosis, and the optional overlay draws the mask boundary onto the original Vacuum-Assisted Resin Transfer Molding (VARTM) frame for human review. Figure Fig. 1 shows the fourteen stages in order.

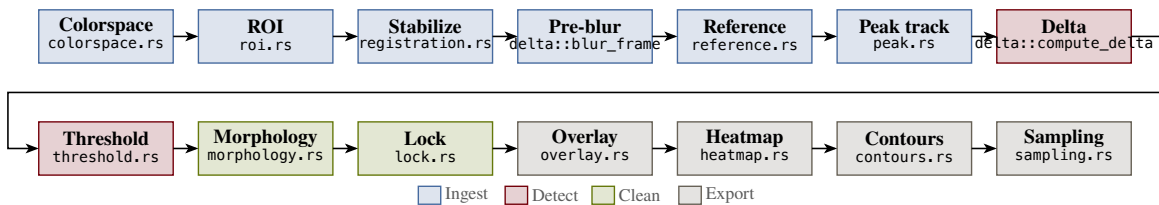


Fig. 1 Fourteen-stage VRIFA pipeline. Each frame flows from decode through an optional camera-shift registration and pre-delta blur, then through an appearance-difference comparison against a chosen reference, into a thresholded and morphologically cleaned mask, and finally through a temporal lock that stabilizes the boundary across frames. The same mask drives the binary, heatmap, and overlay renderers.

The pipeline is deliberately classical, which is what makes it suitable as a reference implementation for downstream learning systems. Every stage is an explicit function of one frame, the chosen reference image, and a small set of scalar parameters, so the output is reproducible bit-for-bit at the stage boundary and the algorithm can be audited end-to-end without recourse to a black-box model.

B. Frame decode and colorspace conversion

The first stage decodes each Blue-Green-Red (BGR) frame from the input video. The second converts that frame into a working colorspace. VRIFA exposes four options, namely the International Commission on Illumination (CIE) 1976 $L^*a^*b^*$ colorspace (CIELAB), Red-Green-Blue (RGB), Hue-Saturation-Value (HSV), and 8-bit grayscale, all routed through standard color converters. CIELAB is the default because resin wetting darkens the lightness channel L^* much more than it shifts chrominance, which makes the single-channel L^* projection used in the difference computation both sufficient and robust. We denote the converted frame at index t by $F_t \in \mathbb{R}^{H \times W \times C}$, where C is the channel count of the chosen colorspace.

C. Region of interest

A rectangular region of interest excludes the part frame, manifold flange, and bag wrinkles outside the laminate. The user supplies four fractional margins, one per edge, each clamped to the closed interval from zero to forty-nine hundredths of the corresponding side, and the algorithm builds a binary mask R that is one inside the resulting rectangle and zero elsewhere. A single configuration parameter sets all four margins symmetrically, with per-edge overrides available. All subsequent stages operate only on pixels where $R = 1$.

D. Camera-shift detection and registration

A bumped tripod, a thermal expansion of the rig, or a hand brushing the camera in mid-run shifts the projected position of every laminate pixel by some vector that has nothing to do with wetting. The reference frame stops aligning with the live frame, the difference field lights up along high-contrast laminate edges, and the threshold catches that as wet. VRIFA detects the shift and corrects it before the difference is computed. For each frame, a single phase-correlation step on a fixed-resolution downsample of the working channel of the previous and current frames returns a translation (d_x, d_y) and a confidence score. When either the per-frame magnitude $\sqrt{d_x^2 + d_y^2}$ or the cumulative drift since the last reset exceeds an operator-set threshold, an iterative-coplanar-correlation refinement fits a translation or affine warp W_t on a static-edge mask of the current ROI, and the live frame is warped through W_t into the reference coordinate system before all downstream stages. The static-edge mask is recomputed at each shift event so the registration is driven by mold and frame edges that do not move with the wet front rather than by the wet region itself. The peak map is reset on registration so the post-warp pixels do not accumulate against pre-warp brightness.

E. Peak-brightness reference

VRIFA accumulates a running maximum of the working channel across all frames seen so far. When the pre-delta blur is enabled the maximum is updated from the blurred working channel rather than the raw channel so the peak map and the delta input share the same bandwidth, which prevents a one-pixel halo of speckle from shifting the peak above the eventual delta input and creating a phantom darkening signal. The motivation for the running maximum is that the dry preform is, by definition, the brightest the fabric will ever appear in the working channel. Treating that running maximum as the per-pixel reference instead of a single early frame absorbs the slow lighting drift caused by lamp warm-up and bag deformation, leaving the difference field free to respond to wetting alone. Figure Fig. 2 shows the effect at one tracked pixel of the canonical input clip. The raw L^* value drifts upward as the lamps stabilize, the running peak P tracks that drift monotonically, and the front arrival cleanly separates as a sharp drop of more than thirty units below the peak. A fixed reference would have started at the value reached on frame zero and missed the post-warm-up lift entirely.

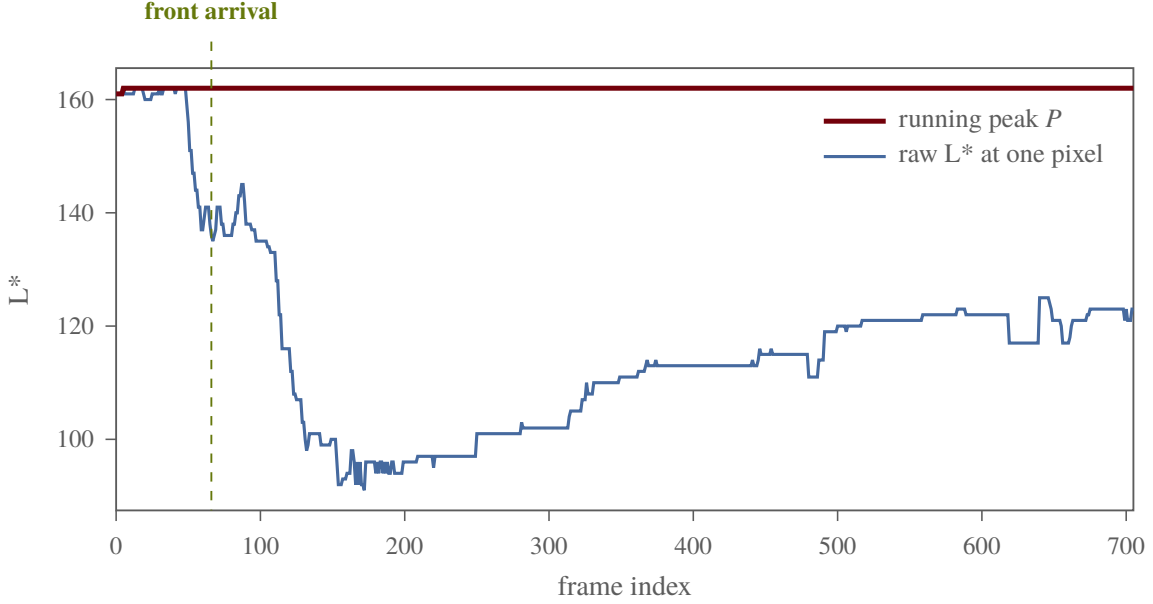


Fig. 2 One pixel of the first canonical input clip tracked across all 706 frames. The raw CIELAB lightness L^* drifts up by roughly twenty units before the front arrives, and the running peak P absorbs that drift. When the resin reaches the pixel, L^* collapses by more than thirty units below the peak, which is the signal the difference stage detects. A single fixed reference at frame zero would have treated the warm-up brightening as a (negative) wetting event.

The peak map can be disabled through a single configuration parameter for users who want a strictly fixed reference, or who are diagnosing a lighting failure where the assumption of monotone brightness no longer holds.

F. Reference selection

The reference image G_t used for the difference computation has five modes. The first-frame mode pins $G_t = F_0$ for every t . The running mode updates an exponential moving average $G_t = (1 - \alpha)G_{t-1} + \alpha F_t$ with default $\alpha = 0.05$. The previous-frame mode uses a fixed-offset history $G_t = F_{t-k}$, with the buffer bootstrapped from F_0 until $t > k$. The absolute mode pins G_t to a user-specified absolute frame index. The dynamic mode adapts the lag online from a square-root-area growth model fit to the early frames. The default selection is the first-frame mode, which, combined with the peak map above, gives a piecewise-static reference whose only adaptation is the peak update.

The dynamic mode warrants a closer look because it is what enables comparable behavior across runs of different fill speeds. For the first N_{calib} frames after detection bootstrapping, VRIFA records the wet area a_t inside the region of interest and the elapsed time $\tau_t = \frac{t-1}{f}$ in seconds, where f is the video frame rate. It then estimates a growth-rate factor κ as the median of $\frac{a_t}{\tau_t^{3/2}}$ across the calibration frames; the three-halves exponent comes from the area-growth law observed for radial Darcy infusion, where wetted area is approximately proportional to time at the three-halves power once flow is well established. Given κ and a target wet fraction ρ of the region-of-interest area $|R|$ (default 0.2), the dynamic lag $\Delta\tau_t$ in seconds that places the reference at the moment when the wet area was a fraction ρ of the current area is

$$\Delta\tau_t = \lambda \cdot \left[\left(\frac{\rho |R|}{\kappa} + \sqrt{\tau_t} \right)^2 - \tau_t \right], \quad (1)$$

clipped to be non-negative and scaled by a user lag factor λ (default 1.0). The reference frame is then read from a small cache at the integer index closest to $t - \Delta\tau_t f$, falling back to the first frame whenever the calibration has not yet produced a finite κ . A linear-mode override is available for diagnostic comparisons.

G. Delta computation

Stage eight produces the per-pixel scalar field D_t that drives all downstream decisions. The default darken-only mode operates on the working (first) channel and records how much darker the frame is than its reference,

$$D_t(y, x) = R(y, x) \cdot \max(0, w_0 \cdot (G_t^*(y, x) - F_t(y, x, 0))), \quad (2)$$

where G_t^* equals the peak map P_t when the peak-reference mode is enabled and otherwise equals the channel-zero slice of G_t . The clip to non-negative values discards every pixel that becomes brighter than the reference, which removes specular flashes from the silicone vacuum bag and from condensation, neither of which are wetting events. A full-color mode replaces the difference with the channel-weighted Euclidean distance across all C channels, exposed through a single configuration parameter and intended for HSV and RGB workflows where chrominance shifts are diagnostic. Figure Fig. 3 shows the effect of the clip.



Fig. 3 Same input frame, two delta computations. The naive Euclidean delta (centre) lights up bright on the vacuum-bag specular highlight to the upper right of the laminate. The darken-only delta (right) discards the highlight and isolates the front. The early-fill front in the lower part of the panel is visible in both, though more cleanly in darken-only.

H. Smoothing, normalization, and threshold

The delta field is smoothed with a separable Gaussian kernel of size $k_b \times k_b$ pixels (default 9, forced odd at runtime) so that the dynamic range of the downstream normalized image is set by structure rather than by isolated speckle. The smoothed field is then rescaled to the byte range using a min-max linear rescaling, producing a $\tilde{D}_t$ that fits in eight bits and feeds both the threshold-selection stage and the heatmap renderer.

Three thresholding modes share a single offset. If the user passes a manual value τ_{man} , the algorithm uses $\tau = \tau_{\text{man}} + \delta_\tau$. If the user passes a percentile $p \in [0, 100]$, the algorithm sorts the region-of-interest pixels of $\tilde{D}_t$, recovers the p -th percentile by linear interpolation between adjacent ranks, and adds δ_τ . Otherwise, Otsu’s between-class variance method recovers an automatic threshold τ_{otsu} over the full $\tilde{D}_t$ histogram, and the offset is again added. In all three cases the final threshold is clipped to the byte range. The default offset $\delta_\tau = -30$ biases Otsu slightly toward the wet class to capture the partially-wetted halo that classical Otsu would otherwise miss; manual and percentile modes are reserved for parameter studies and are not used for the headline numbers reported in this paper.

I. Mask cleanup

The binary mask leaving the threshold stage is rough. It contains gaps where transient bag wrinkles muted the local response, isolated specks where the threshold caught a single noisy pixel, and components small enough that they are obviously noise rather than wetted fabric. The morphological cleanup and the connected-components filter remove those artefacts in a fixed sequence. Morphological closing with an elliptical structuring element of size k_m (default thirteen pixels) fills the gaps and welds neighbouring patches into one front. Morphological opening with the same kernel removes specks below the kernel size. Connected-components labelling discards any region whose pixel area is below a_{min} pixels (default 400). Figure Fig. 4 shows the same frame at every step of that sequence.

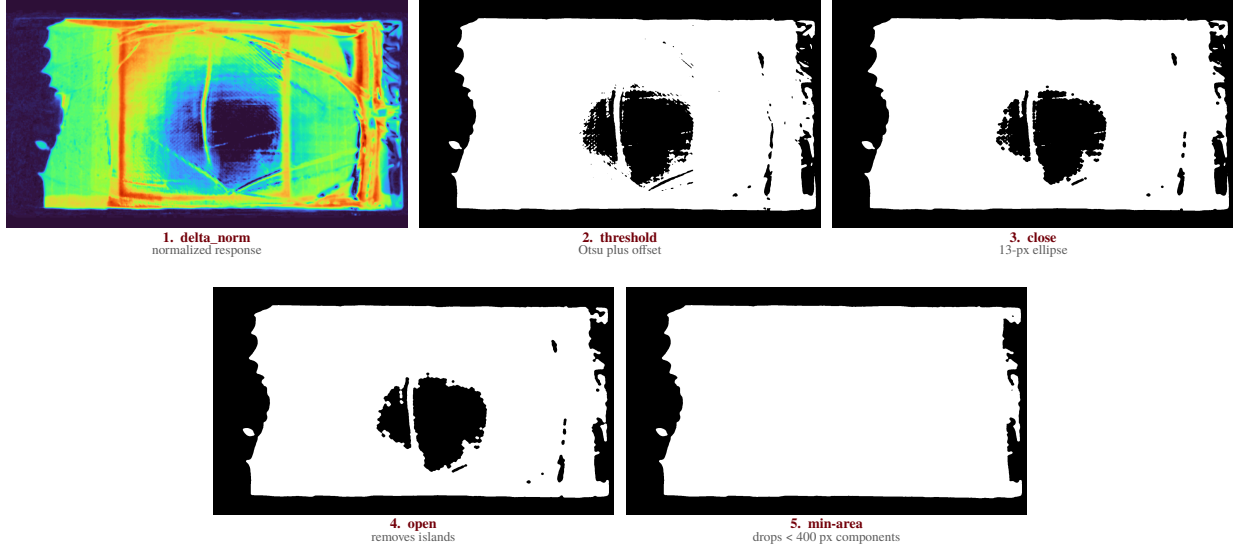


Fig. 4 Mask cleanup on the first canonical input clip at frame 200. The normalized response field (1) is thresholded into a noisy binary mask (2). Closing welds neighbouring wet patches and fills small gaps (3). Opening removes specks the closing kernel could not fill (4). Connected-components labelling removes the few residual islands below the four-hundred-pixel area floor, leaving the final mask on which the temporal lock operates (5). The kernel parities are forced odd at runtime so that anchor handling is symmetric. The structuring-element shape is elliptical by default with optional rectangular and cross-shaped alternatives, exposed because the front shape changes with infusion geometry.

J. Temporal locking

Stage thirteen imposes hysteresis along the time axis. Each pixel keeps a small per-pixel counter that increments while the cleaned mask reports the pixel as wet and resets to zero on any frame where the cleaned mask says dry. Once the counter reaches the threshold n_{lock} frames, a sticky locked-pixel map is set true at that location and never resets. The output of the stage is the elementwise OR of the cleaned mask with the sticky locked map, so a pixel that has ever been wet for n_{lock} consecutive frames stays wet for the remainder of the run. Figure Fig. 5 illustrates the bookkeeping with a twelve-frame example for $n_{\text{lock}} = 3$.

per-pixel state by frame

	1	2	3	4	5	6	7	8	9	10	11	12
detection	0	0	1	1	0	1	1	1	1	0	1	1
counter	0	0	1	2	0	1	2	3	4	0	1	2
locked?	×	×	×	×	×	×	×	✓	✓	✓	✓	✓

latches at frame 8

Fig. 5 Lock state for one pixel across twelve frames with $n_{\text{lock}} = 3$. The detection row records what the cleaned mask said this frame. The counter row tracks consecutive wet frames and resets on dry. The locked row latches at the first frame where the counter reaches three (frame eight in this example) and never returns to false. A flicker that survives for fewer than three frames is treated as a transient and not committed.

The default $n_{\text{lock}} = 3$ trades a small number of frames of recovery latency for boundary stability that downstream consumers need. Setting the lock window to zero disables the stage altogether for users who want raw per-frame masks.

K. Heatmap, overlay, and contour export

The last two display stages exist for inspection and label export. The heatmap renderer maps the normalized delta $\tilde{D}_t$ through the Turbo colormap to produce a three-channel pseudocolor image. The overlay renderer extracts the boundary of the locked mask via a five-by-five rectangular morphological gradient and paints those boundary pixels red on a copy of the original BGR frame. Neither renderer is part of the detection logic; they expose, in human-readable form, the field on which the threshold acted and the boundary that the locked mask encloses.

For machine-readable export, the contour-extraction stage emits Common Objects in Context (COCO) and YOLO-format polygons for every connected component of the locked mask. Polygon segmentation is computed with a standard contour-extraction routine, optionally simplified by the Douglas-Peucker algorithm with tolerance ε , and optionally densified to a maximum edge length so that downstream rasterization preserves curvature. The annotation-sampling utility selects which frames receive labels using one of three modes, namely all-frame, evenly-spaced count, or fixed-stride, with deduplication of consecutive ties so that the integer-truncated linear-spacing exactly reproduces the standard reference behaviour.

L. Configuration values held fixed across all experiments

Table Table 1 lists the configuration values held fixed across every experiment reported in this paper, including the per-component ablation. The values were chosen during development on a held-out subset disjoint from the labeled evaluation set, were not retuned per video or per metric, and are reproduced here so that any reader can recover the exact operating point of every reported number. The ablation in Section 5 reports what happens when each row is changed in isolation.

Table 1 Configuration values held fixed across every experiment reported in this paper. Symbols match the variables introduced in the preceding subsections. The only equation that depends on them explicitly is the dynamic-mode lag of Eq. Eq. (1). The component-removal ablation in Section 5 reports the IoU effect of changing each row in isolation.

Symbol / parameter	Default	Description
colorspace	CIELAB	Working colorspace for the difference computation.
roi-margin	0.15	Symmetric fractional ROI margin per edge.
ref-mode	first	Reference selection mode.
ref-running-alpha, α	0.05	EMA weight for the running reference.
peak-reference	true	Use the running peak map in the working channel.
darken-only	true	Clip the delta to non-negative wetting deltas.
camera-stable	false	Enable phase-correlation camera-shift detection.
motion-per-frame-threshold	1.5	Per-frame translation magnitude that triggers a registration.
cumulative-motion-threshold	3	Accumulated drift, in pixels, that triggers a registration.
motion-model	affine	Warp model fit on a static-edge mask when a shift is detected.
peak-on-shift	reset	How the peak map is treated when a shift is registered.
pre-delta-blur-kernel, k_p	0	Gaussian blur applied to the working frame before peak update and delta. Disabled when zero.
blur-kernel, k_b	9	Gaussian blur kernel size in pixels, applied to the delta field.
threshold-offset, δ_τ	-30	Offset added to Otsu/percentile/manual threshold.
morph-kernel, k_m	13	Morphology structuring-element size.
morph-shape	ellipse	Structuring-element shape.
morph-close-iterations	1	Morphological closing iterations.
morph-open-iterations	1	Morphological opening iterations.
min-area, $a_{\min}$	400	Minimum connected-component area, in pixels.
lock-frames, n_{lock}	3	Temporal lock window, in frames.
frame-step	1	Frame stride at decode time.
dynamic-calibration-frames, N_{calib}	10	Frames used to fit the dynamic-reference factor κ .
dynamic-target-fraction, ρ	0.2	Target wet-area fraction for dynamic-reference lag.
dynamic-lag-scale, λ	1.0	Multiplicative scale on the dynamic-mode lag.
dynamic-ref-cache-size	32	Frames cached for the dynamic-reference reader.

V. Datasets and Labeling Protocol

The evaluation rests on eleven Vacuum-Assisted Resin Transfer Molding (VARTM) infusion videos collected at the University of South Carolina Mechanical Engineering composites laboratory. Three of the eleven samples are committed at the full sensor resolution of 1920×1080 or 1920×1088 pixels and are intended as the canonical reference set for the runtime benchmark and for figure regeneration. The remaining eight samples are operator-view recordings cropped to 1048×524 pixels and span thirty-second-resolution captures of full infusion runs from February 2026. Sample durations range from twenty-three seconds for the shortest standard-rate clip to roughly nine minutes for the longest cropped run, and frame counts range from one hundred to over fifteen thousand. Two of the high-resolution clips are recorded at a time-lapse rate of 3.33 frames per second, while the remaining nine record at 30 frames per second. The sample inventory is summarized in Table Table 2.

Table 2 Sample inventory. Eleven VARTM infusion clips committed under `data/`. The high-resolution clips serve as the canonical benchmark targets; the cropped operator-view recordings contribute the long-form generalization evidence.

Sample	Frames	FPS	Width	Height	Class
input_1	706	29.97	1920	1080	high-res standard
input_2	100	3.33	1920	1088	high-res time-lapse
input_3	200	3.33	1920	1088	high-res time-lapse
input_4	8 100	30.00	1048	524	cropped operator
input_5	8 100	30.00	1048	524	cropped operator
input_6	8 100	30.00	1048	524	cropped operator
input_7	8 100	30.00	1048	524	cropped operator
input_8	12 600	30.00	1048	524	cropped operator
input_9	15 469	30.00	1048	524	cropped operator
input_10	11 426	30.00	1048	524	cropped operator
input_11	14 876	30.00	1048	524	cropped operator

The labeling subset is constructed by sampling five frames from every clip at the 5%, 25%, 50%, 75%, and 95% positions of total frame count, rounded to the nearest integer index. The sampling is stratified within each sample so that the fifty-five-frame ground-truth set covers every clip equally rather than concentrating in a single high-frame-count recording. The resulting set spans early-fill, early-mid, mid, mid-late, and late-fill stages within each clip, and across clips it spans the four resolution and frame-rate regimes described above.

A. Labeling protocol

Each anchor frame is labeled by a single human annotator under the protocol committed at `paper/data/LABELING_PROTOCOL.md`. The annotator marks one polygon per frame indicating the resin-wetted region, defined as the area where resin has visibly transitioned the fabric from dry to saturated. Specular reflections from the silicone vacuum bag, vacuum-bag wrinkles and creases that pre-date the front, fabric-weave shadows that pre-date the front, and pooled resin in inlet runners outside the laminate boundary are explicitly excluded. The annotator distinguishes real wetting from transient appearance changes by scrubbing forward a few frames and confirming that the candidate region’s brightness has changed monotonically. Real wetting darkens once and stays dark; reflections and wrinkles oscillate. The labeling tool is the browser-based polygon editor at `makesense.ai`, chosen for its ability to import a flat directory of PNGs and export Common Objects in Context (COCO) JSON without intermediate format conversion. Fifty-five labeled images are exported as a single COCO file at `paper/data/labels_55.json`. Figure Fig. 6 shows a representative labeled frame.

Labeling protocol illustration placeholder. One representative frame from the fifty-five-frame ground-truth subset, with the operator’s polygon overlaid in rose. Replaced once labels arrive.

Fig. 6 Example labeled frame from the labeling pass. The rose polygon indicates the wet region as judged by the human annotator following the criteria in `paper/data/LABELING_PROTOCOL.md`.

B. Scope and known limits of the labeling subset

A subset of fifty-five frames is small in absolute terms and reviewers should treat per-sample agreement numbers as first-pass evidence rather than as definitive benchmarks. The strategy here is breadth over depth, stratifying across all eleven clips so the per-sample breakdown surfaces any sample-specific mode of failure rather than concentrating evidence on a single recording. The agreement script tolerates partial label sets, and scaling the labeled subset to one hundred and ten frames (ten per sample) is a straightforward extension if a reviewer demands more. The locked thesis sentence in this paper accommodates that scaling without rewording beyond the count.

A single human annotator was used. Inter-annotator agreement is not reported. The protocol was written before the labeling pass began, however, so the criteria for what counts as wet are recoverable and could be applied by a second annotator in follow-on work without ambiguity. The protocol document is committed alongside the labels to support exactly that kind of replication.

VI. Quantitative Agreement Results

The agreement between the VRIFA mask and the human-labeled ground truth is measured frame-by-frame on the fifty-five-frame subset described in Section 5. For every labeled frame, the human polygon is rasterized into a binary mask at the source resolution and compared against the mask produced by the default configuration of the Rust binary. Five complementary metrics are reported. Mask Intersection-over-Union (IoU) is the headline accuracy metric. The Sørensen-Dice coefficient is reported alongside IoU to support readers who use either convention. Boundary F_1 is computed at the mean of three pixel tolerances $\tau \in \{1, 3, 5\}$ pixels, where the per-tolerance F_1 is the harmonic mean of precision (prediction-boundary pixels within τ pixels of any ground-truth-boundary pixel) and recall (ground-truth-boundary pixels within τ pixels of any prediction-boundary pixel). Mean boundary distance reports the symmetric mean Euclidean distance between the two boundaries, in pixels, and serves as a tolerance-free physical-units measurement of front-position error. Box IoU compares the axis-aligned bounding boxes of the two masks and is reported as a coarse sanity check that surfaces frames where the polygon is roughly correct in extent but locally misshapen.

Bootstrap 95 % confidence intervals are reported for every metric, computed from 10,000 resamples of the per-frame metric values with replacement. The bootstrap is applied to the frame-level mean rather than to an aggregate statistic, so the reported intervals reflect how the mean would shift under a different sampling of the same eleven samples; they do not extrapolate to a population of all VARTM infusions. The script that produces these numbers is committed at `_dev/validation/agreement.py`, and the resulting machine-readable JSON is committed at `paper/data/agreement_metrics.json`.

A. Overall agreement

Table 3 Overall agreement across the fifty-five-frame labeling subset. Confidence intervals are bootstrap quantiles over ten-thousand resamples of the per-frame mean.

Metric	Mean	95 % CI
Mask IoU	<i>TBD</i>	[<i>TBD</i> , <i>TBD</i>]
Sørensen-Dice	<i>TBD</i>	[<i>TBD</i> , <i>TBD</i>]
Boundary F_1	<i>TBD</i>	[<i>TBD</i> , <i>TBD</i>]
Mean boundary distance (px)	<i>TBD</i>	[<i>TBD</i> , <i>TBD</i>]
Box IoU	<i>TBD</i>	[<i>TBD</i> , <i>TBD</i>]

B. Per-sample breakdown

The eleven samples described in Section 5 differ substantially in resolution, frame rate, illumination, and operator framing, and the per-sample breakdown is the strongest available evidence that the agreement reported above is consistent across substantively different molds rather than driven by a single fortunate recording. Table Table 4 reports mask IoU and boundary F_1 for each sample alongside the count of labeled frames contributing to the mean.

Table 4 Per-sample agreement for mask IoU and boundary F_1 . Each row aggregates the five anchor frames sampled at the $\frac{5}{25}, \frac{50}{50}, \frac{75}{75}, \frac{90}{90}$ % fill positions described in Section 5.

Sample	n	Mask IoU	Boundary F_1
input_1	5	<i>TBD</i>	<i>TBD</i>
input_2	5	<i>TBD</i>	<i>TBD</i>
input_3	5	<i>TBD</i>	<i>TBD</i>
2026-02-17_...	5	<i>TBD</i>	<i>TBD</i>
2026-02-18_...	5	<i>TBD</i>	<i>TBD</i>
2026-02-19_...	5	<i>TBD</i>	<i>TBD</i>
2026-02-20_...	5	<i>TBD</i>	<i>TBD</i>
2026-02-23_...	5	<i>TBD</i>	<i>TBD</i>
2026-02-24_...	5	<i>TBD</i>	<i>TBD</i>
2026-02-25_...	5	<i>TBD</i>	<i>TBD</i>
2026-02-26_...	5	<i>TBD</i>	<i>TBD</i>

C. Qualitative montage

Frame montage placeholder. Two rows of three row / overlay pairs: input_1 frames 50, 200, 500 (top); input_2 frames 30, 60, 90 (bottom). Plus three labeled-vs-predicted comparisons drawn from cropped operator-view samples.

Fig. 7 Frame montage at the parity-validated anchor frames. Top two rows show raw input alongside the VRIFA overlay for the high- resolution input_1 and input_2 reference videos. Bottom row shows three labeled-versus-predicted comparisons drawn from the cropped operator-view samples to surface failure modes that the high-resolution videos do not exhibit.

VII. Runtime and Systems

The integrated pipeline runs at video rate on a single CPU thread, and the same algorithm has been ported to a CUDA implementation that processes the same eleven samples at substantially higher throughput. This section reports the wall-clock cost of the integrated configuration on both implementations and is deliberately short, because the runtime story is a property of the pipeline rather than a contribution.

A. Hardware

All runtimes are reported on a single host. The CPU is an Apple M-series with N_{cores} performance cores at f_{cpu} GHz, N_{ram} GB unified memory, running macOS V_{os} . The GPU runtimes use an NVIDIA G_{model} with V_{vram} GB of

memory, driver version V_{drv} . Decode and encode use the system FFmpeg with the libx264 and libopenh264 codecs, both invoked through the OpenCV `videoio` interface. No frame is decoded twice and no result is cached between runs.

B. Throughput

Table Fig. 8 reports per-sample wall-clock for the integrated configuration on the CPU implementation, on the CUDA implementation, and on a Python reference implementation that mirrors the algorithm stage-for-stage and is included as a runtime baseline rather than as a science contribution. Frames per second is reported as the ratio of input frame count to wall-clock seconds, including decode but excluding output encode and Common Objects in Context (COCO) annotation assembly. The CUDA implementation reaches K frames per second aggregated across the eleven samples, an S -fold speedup over the CPU implementation at the same parameter values.

Three-implementation runtime placeholder. Aggregate per-sample wall-clock and frames per second for the Python reference, the CPU implementation, and the CUDA implementation across the eleven VARTM samples. Replaced once the CUDA close-out completes.

Fig. 8 Per-sample wall-clock for the integrated configuration on three implementations of the same algorithm. Wall-clock includes video decode but excludes output encode and annotation export. Speedups are reported relative to the Python reference at matched parameter values.

C. Validation

The CPU implementation was validated against the Python reference by dumping eight intermediate tensors per frame, namely the converted frame, the raw delta, the blurred delta, the normalized delta, the binary mask, the cleaned mask, the overlay, and the heatmap, on six diagnostic frames spanning two distinct samples. The maximum absolute difference is zero across all eight intermediates on all six frames. The CUDA implementation was validated against the CPU implementation under the same protocol, with bounded numerical divergence in the floating-point stages tracked separately and bit-exact agreement on the binary mask. The validation confirms that the runtime numbers in Table Fig. 8 correspond to the same algorithm at all three operating points, not to three different algorithms that happen to agree on summary statistics.

VIII. Discussion

VRIFA is a working baseline, not a universal segmenter. The failure modes described below are concrete, mechanism-level, and visible from the operator console, which is the same place the central processing unit (CPU) detection pipeline runs in production.

A. Failure Modes

The temporal-locking parameter is the primary diagnostic when results appear noisy on long pauses. The pipeline exposes a temporal-locking window, defaulting to three frames. Locking holds a positive detection in the mask for that many subsequent frames so transient single-frame dropouts do not flicker the boundary. When the operator sets the temporal-locking window higher than the expected pause length in the run, transient detections from earlier in the infusion persist past the moment the front recedes, and the mask grows phantom regions that look like artifacts. The mechanism is not a defect. It is the locking semantics doing exactly what they were configured to do, applied to a sequence whose pauses violate the implicit assumption. The operational consequence is that the temporal-locking window is a per-mold parameter, not a global default, and the failure is visible only on long-pause sequences.

The dynamic-reference calibration window is the second sensitive surface. The pipeline exposes a dynamic-calibration-frames parameter, defaulting to ten, and the dynamic reference mode uses those first ten processed frames to

fit a square-root-of-area growth model that subsequently lags the reference frame behind the live frame. The assumption baked into that fit is that early-fill is representative of the growth law. If the first ten processed frames contain race-tracking, an air pocket, or some other anomaly, the fit picks up a poor reference factor and every downstream frame inherits that error. The fix is operator-side, namely choose a calibration window that excludes the anomaly or fall back to one of the static reference modes. The failure is again a misuse of a parameter rather than a numerical defect.

The third failure mode is the inherent ceiling of any tuned classical-CV pipeline. The integrated configuration does not generalize across substantially different fabric types, lighting setups, or vacuum-bag textures without re-tuning. The configuration values in Table 1 assume a transparent vacuum bag, top-down LED lighting, and a dark mold background, because those are the conditions of the eleven samples in the labeling subset. Off-distribution conditions, such as a heavily textured bag, side lighting, or a bright mold surface, can break the Otsu threshold or push the response distribution outside the range the threshold offset was tuned for. The pipeline exposes a percentile-threshold mode, a manual contrast threshold, and an RGB colorspace switch precisely because no single configuration covers every regime, and a per-mold reconfiguration is the appropriate response. The component-removal ablation in Section 5 reports on conditions where the integrated configuration as evaluated holds.

B. Tradeoffs

VRIFA is built on a deliberate set of tradeoffs that a reader should weigh alongside the benchmark numbers.

We trade an end-to-end accuracy ceiling for explainability and per-mold tunability. A learned segmenter trained on a sufficiently large in-domain corpus would likely exceed the boundary F1 reported in this paper. We do not have a sufficiently large in-domain corpus, and a research group that wants to run a vacuum-assisted resin transfer molding (VARTM) experiment next week does not have the time to label one. The Application contribution is precisely the claim that this tradeoff is favorable in the VARTM regime, where samples are small, ground truth is sparse, and the operator already understands the physics. The pipeline runs on a CPU with no graphics processing unit (GPU), which keeps the deployment surface small and matches how VARTM rigs are actually instrumented in academic and industrial labs.

We trade automated hyperparameter selection for human inspection. There is no Optuna-driven default and no opaque end-to-end model. The operator sees what each parameter does. The roughly fifty configuration parameters are a feature for research, where the practitioner needs to know exactly why a particular result looks the way it does, and a hazard in production, where an unfamiliar operator can choose a temporal-locking window that ruins a run. We are explicit about that tension rather than pretending the parameters are zero-cost.

We trade end-to-end byte-exactness across runs for a working video output. The MP4 streams are not bit-reproducible, and we report rather than hide the encode-tier noise. Bit-reproducibility is preserved where it matters for downstream supervision, namely on the PNG outputs and on the COCO annotations, and abandoned where it does not, namely on the human-review video.

We trade end-to-end accuracy for explainability of every component’s effect. The component-removal ablation in Section 5 reports the IoU consequence of disabling each named primitive, which is the price of the integrated configuration framing. A monolithic learned segmenter would not expose those deltas, so a reader interested only in the headline number could replace this pipeline with a sufficiently large convolutional model and lose the per-component evidence. The contribution is the evidence, not the headline.

C. Scope and Limits

Several setups fall outside the regime VRIFA targets. Opaque vacuum bags break the entire pipeline, because the front is no longer visually observable from above and there is nothing for any image-domain method to detect. Multi-resin systems where dye contrast is intentionally absent fall in the same category, since the response image VRIFA computes is, ultimately, a contrast measurement. Very fast infusions, where the front passes through a region of interest in well under a second, push the temporal-locking and morphology kernels past the regime they were designed for, and the resulting masks lag the true front. Camera motion during infusion is handled by the optional registration stage, which warps the live frame back into the reference coordinate system whenever per-frame or cumulative drift exceeds an operator-set threshold. The registration absorbs the bumped-tripod and thermal-creep regimes that previously broke the comparison between live and reference frame; setups where the camera moves continuously and at high amplitude,

such as a handheld phone walked around the rig, exceed the small-shift assumption the affine warp is fit under and remain out of scope.

These are not weaknesses to be patched. They are the boundary of the application domain. A practitioner facing any of them requires a different tool, and stating that boundary explicitly is more useful than overstating the method's operating envelope.

D. Implications for the Field

The integrated pipeline is intended to be useful to two audiences. A practitioner running a VARTM rig obtains a flow-front segmentation method whose effect on each captured infusion is interpretable component by component, so a per-mold problem (specular bag, off-distribution lighting, fast-fill geometry) can be traced to a specific stage rather than buried in an end-to-end loss. A researcher building permeability-inversion or process-monitoring studies that depend on per-pixel front geometry obtains a method whose individual components are characterized against a labeled benchmark, so substituting any one of them for an alternative method can be evaluated against a fixed comparison point rather than against an opaque baseline.

IX. Conclusion

The integrated pipeline reaches mask IoU $X.XXX$ and boundary F_1 $Y.YYY$ across a 55-frame hand-labeled subset spanning eleven distinct VARTM infusion runs. Removing any single component, namely the peak-brightness reference, the darken-only delta, the region-of-interest restriction, the dynamic-lag reference selection, the morphological cleanup, or the temporal lock, drops mean IoU by at least ΔIoU absolute on the same subset. The pipeline runs at 30 frames per second on a single CPU. The empirical contribution is therefore not any one of the primitives in isolation, all of which already appear in prior VARTM and Liquid Composite Molding work, but the joint configuration and the per-component evidence that the joint matters.

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